Tianye Zhang

Ann Arbor, MI
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EDUCATION

University of Michigan, Ann Arbor, MI **Graduation Date (05/ 2022)**
Mechanical Engineering BSE GPA: 3.82 /4.0

- Honors/Awards: University Honors (2019-2022), Dean's List (2019-2022), James B. Angell Scholar (2020-2022)
- Undergraduate IA in ME240 Introduction to Dynamics and Vibrations, GSI in ME395 Lab I
- Grader in ME311 Strength of Materials, ME424 Engineering Acoustics
- Course Highlights: Mechanics, Design, Acoustics

University of Michigan, Ann Arbor, MI **Graduation Date (05/2024)**
Mechanical Engineering MSE / Biomechanics GPA: 3.72 /4.0

- Course Highlights: Medical Imaging, Advanced Engineering Acoustics

University of Michigan, Ann Arbor, MI
Mechanical Engineering PhD Student (2025–present; second year)

RELATED EXPERIENCE

UM Meta Acoustic Lab **Sept. 2024 - Current**
Researching Student Ann Arbor, MI

- Design, simulate and conduct experiments for ultrasound neuromodulation
- Design, build and fabricate ultrasound scanner setup
- Design, synthesis, and fabricate nanoparticles for ultrasound-induced luminescence
- Investigate acoustic emission enhancement and nonlinear dynamics in phase-changing nanodroplets
- Develop models of resonant droplet pressure, nucleation exposure, and weak-field acoustic control

Tang Group **Sept. 2022 - Aug 2024**
Researching Student Ann Arbor, MI

- Material characterization and design for bioelectronics for biomedical application

MIT Zhao Lab **Jun. 2022 - Sept. 2022**
Visiting Student Cambridge, MA

- Designed and performed simulations for hydrogel bio-metamaterials with mechanical flexibility and acoustic gradient refractive index

University of Michigan Sensor Network Laboratory: Mapleseed Project **Sep. 2021 - Jan 2022**
Researching Student Ann Arbor, MI

- Designed and selected mechanism for microparticle visualization on drone-based optical microscope
- Modeled and fabricated plano convex lens setup for microparticle visualization

Xiamen University MBAT Lab **Dec. 2020 - Aug. 2021**
Visiting Student Xiamen, FJ

- Designed and setup the laboratory for soft material design
- Researched and designed bioinspired jellyfish with ultrasound transparency

- Performed modeling, design and fabrication of soft pneumatic generator
- Developed and investigated underlying mechanism for metagel focusing through aberrating layers
- Designed and fabricated bioinspired snapping shrimp device for cavitation-based sound generation

MIT Nanophotonics and 3D Nanomanufacturing Lab **Dec. 2018 - Aug. 2019**
Visiting Student Cambridge, MA
- Conducted research on physical modeling and validation of the bioinspired ultrasound emission via metamaterial

FIRST Robotics Competition (FRC) Team Nutrons (125) **Dec. 2017 - Aug. 2019**
Lead Mechanical Student Boston, MA
- Designed, manufactured and assembled subsystems (telescoping arm and double lift, climber and partner lift) for 2018 and 2019 FRC game
- Won New England District Championships, Division Finalists at Worlds in 2019

SKILLS
- Modeling: Solidworks, Comsol, Rhino, AutoCAD, Simulink, Altium
- Coding: Matlab, C++, Python, HTML, Javascript, Mathematica
- Art: Adobe Photoshop, Adobe Illustrator

PUBLICATIONS († equal contribution)
1. Jinhu Zhang†, **Tianye Zhang**†, Erqian Dong, Chuang Zhang, Zhonglu Lin, Zhongchang Song, Hongquan Li, Nicholas X. Fang, and Yu Zhang. Bioinspired hydrogel jellyfish with mechanical flexibility and acoustic transparency. Cell Reports Physical Science 3(10), 101081 (2022).
2. Erqian Dong†, **Tianye Zhang**†, Jinhu Zhang, Xiaochun Su, Sichao Qu, Xin Ye, Zhanyuan Gao, Chengtian Gao, Jiangang Hui, Zhanxiang Wang, Nicholas X. Fang, and Yu Zhang. Soft metalens for broadband ultrasonic focusing through aberration layers. Nature Communications 16, 308 (2025).
3. Jinhu Zhang†, Deshuai Yu†, **Tianye Zhang**, Chengtian Gao, Erqian Dong, Zhongchang Song, Chaoyu Fan, Zhehao Han, Fabrice Lemoult, Mathias Fink, Zhanxiang Wang, Youhui Lin, and Yu Zhang. Hydrogel metapad with ultrasound transparency and broadband focusing for biomedical imaging. National Science Review 13(6), nwag048 (2026).
4. Erqian Dong, Yu Zhang, Zhongchang Song, **Tianye Zhang**, Chen Cai, and Nicholas X. Fang. Physical modeling and validation of porpoises' directional emission via hybrid metamaterials. National Science Review 6(5), 921–928 (2019).

PREPRINTS AND MANUSCRIPTS
1. **Tianye Zhang**†, John Kim†, Heidi Zhang, Ge Wang, Ke Ding, Yan Deng, Chenzhe Wang, Jianyu Hua, and Chengzhi Shi. Engineering the Nanoscale Acoustic Purcell Effect. Preprint (2025); submitted manuscript in revision (2026). DOI: 10.21203/rs.3.rs-7958211/v1.
2. **Tianye Zhang**, Ge Wang, Chenzhe Wang, Jianyu Hua, and Chengzhi Shi. Acoustic phase control of internal pressure and nucleation exposure in resonant liquid microdroplets. Submitted to The Journal of the Acoustical Society of America (2026).
3. **Tianye Zhang**†, Ge Wang†, Chenzhe Wang, Jianyu Hua, and Chengzhi Shi. Chaotic dynamics and control in a resonance-mediated acoustic droplet vaporization model. Submitted to Nonlinear Dynamics (2026).
4. **Tianye Zhang**, Ge Wang, John Kim, and Chengzhi Shi. Weak-field regulation of phase-change nanodroplets enables acoustic nano-encoding. Manuscript prepared for resubmission (2026).